The Media Connectivity Association



OVERVIEW AND UPDATES · IMMERSIVE MEDIA

Volumetric Video Experiences

Volumetric video platform for immersive experiences

MPEG (ISO/IEC)

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About the project

V3C represents 3D content so that conventional 2D video codecs do the compression

ISO/IEC 23090-5

MPEG standard

Visual volumetric video-based coding (V3C) and video-based point cloud compression (V-PCC).

ISO/IEC 23090-12

MPEG standard

MPEG Immersive Video (MIV), an extension of the V3C base.

ISO/IEC 23090-10

With 3GPP TS 26.512

Carriage of V3C data in ISOBMFF with support for DASH, and the protocols and APIs that deliver V3C DASH presentations over 5G Media Streaming.

FIND OUT MORE **Technology** **Specifications** **Reference Tools**

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Profiles and components

What V3C defines, and what a V3C bitstream carries

V-PCC ISO/IEC 23090-5: a dynamic point cloud, typically one captured object or performer

- Projection onto per-normal planes
- Packing of projections
- Geometry, occupancy and attribute videos

MIV ISO/IEC 23090-12: several cameras with depth, six degrees of freedom in a limited viewing volume

- Main
- Extended
- Extended Restricted Geometry
- Geometry Absent

Bitstream components clearly separated

- Atlas sub-bitstream
- Common atlas sub-bitstream (MIV)
- Occupancy video
- Geometry video
- Attribute video(s)

Carriage ISO/IEC 23090-10

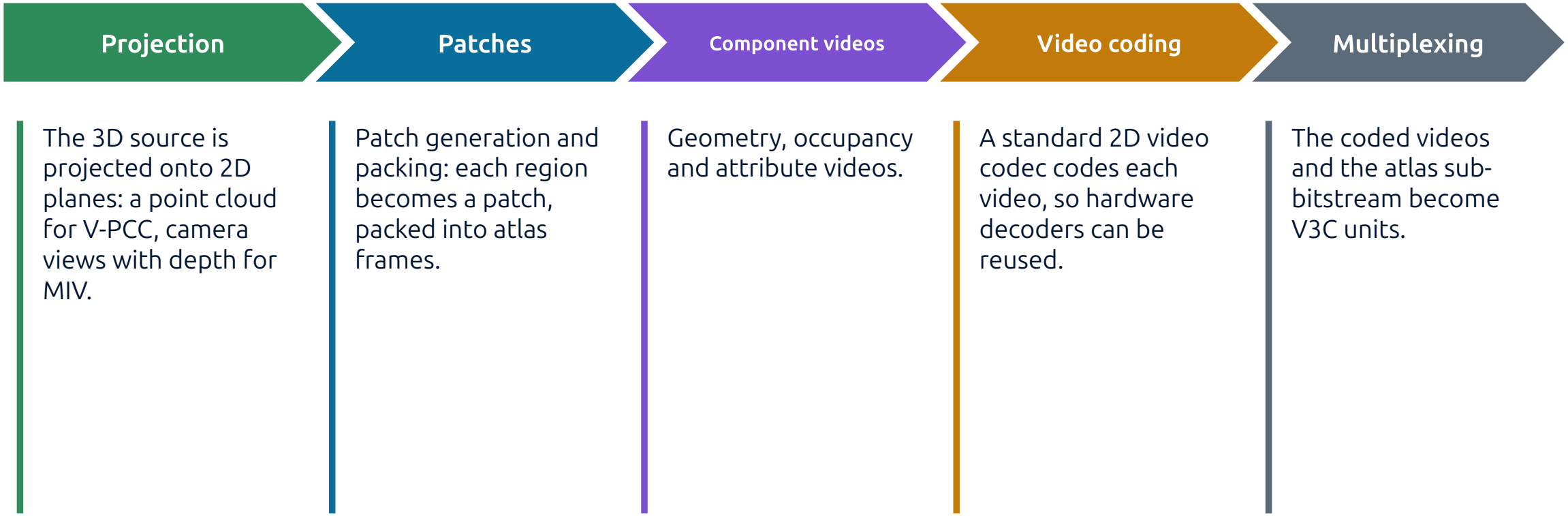
- ISOBMFF
- DASH

The component videos are ordinary coded video, so a decoder can offload them to a hardware video decoder and keep CPU and GPU work for the atlas and the 3D reconstruction.

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How it works

The projection-based pipeline of V3C, from the 3D source to the bitstream

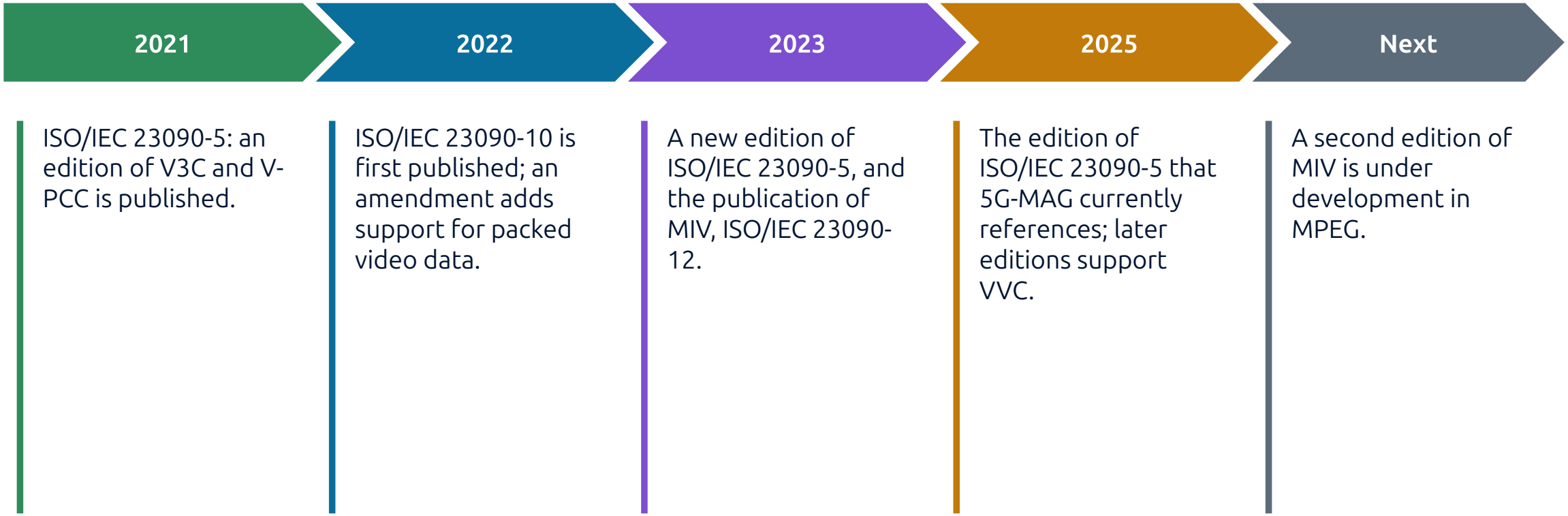


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How it grew

The editions of the MPEG specifications behind the platform



Confirm the exact edition and any amendments in the ISO catalogue for the feature set you rely on.

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Repositories

What each repository covers, from its README; support of editions and profiles is not yet audited

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE	
rt-v3c-unity-player	a Unity package to decode and play V3C content, with a simple demo project	Player, Unity, Windows, Android	v1.1 30 Sep 2025	GitHub
rt-v3c-decoder-plugin	entry points for decoding and viewing MIV and V-PCC content	Decoder, DASH client, Synthesizers	v1.1 30 Sep 2025	GitHub
rt-v3c-content	test content for the Unity player, and tools to generate V3C bitstreams and DASH segments	Test content, DASH packager	v1.1 9 Oct 2025	GitHub
rt-v3c-examples	example V3C content and helper files for testing the encoding, delivery and rendering pipeline	Test content	none yet	GitHub
rt-media-origin	a config-driven media origin that packages DASH, HLS or CMAF and delivers over HTTP or push protocols	Media origin, DASH, CMAF	none yet	GitHub

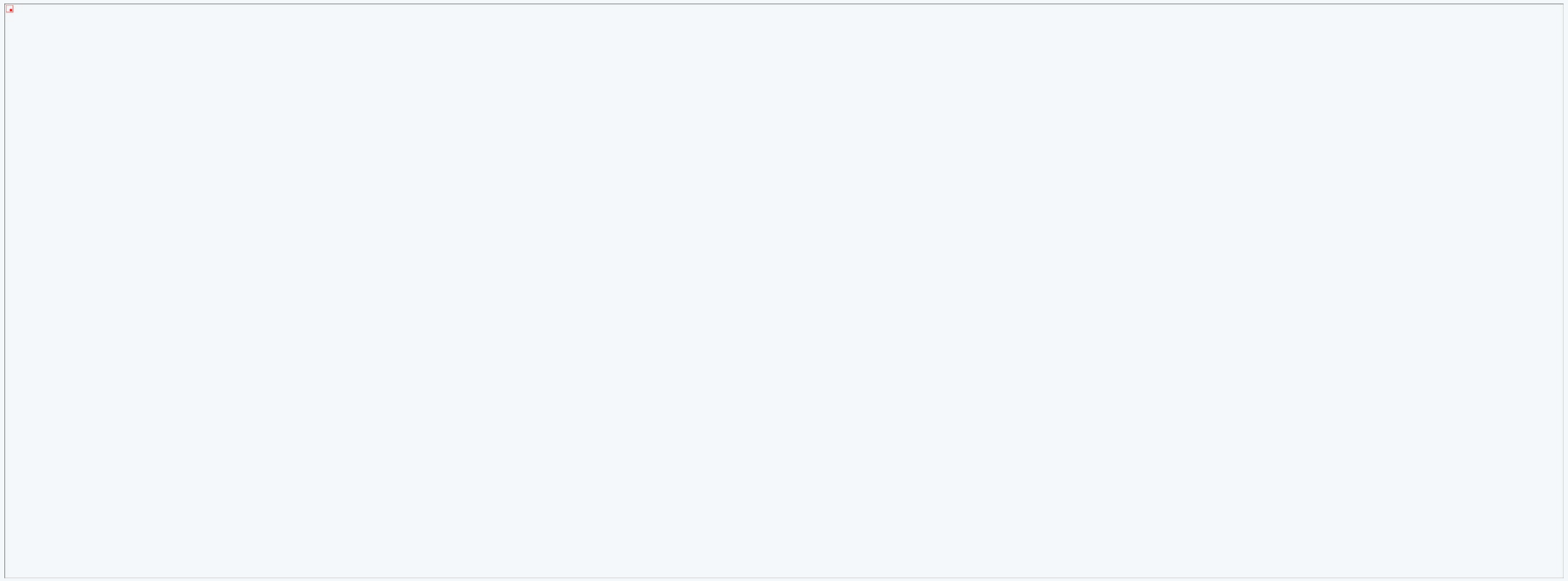
FIND OUT MORE Technology Specifications Reference Tools

● released ● no release yet

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Repositories in the architecture

The Reference Tools on the specification figure, as the figure generator draws them



PROVIDED BY THE REFERENCE TOOLS

1. **rt-v3c-unity-player**
2. **rt-v3c-content**
3. **rt-v3c-decoder-plugin**
   used by 1

Boxes without a badge are not mapped to a repository.

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Build and try

The tutorial: the V3C Unity Player on an Android phone, streaming from a DASH server

Build the plugin	Build the player	Copy the data	Stream
Build the V3C decoder plugin for Android and copy it into the Unity player.	Build the V3C Unity Player for Android: an APK to install on the phone.	Copy on-device-data to the phone and on-server-data to the DASH server.	Stream from a simple express DASH server on your own machine.

The player needs OpenGL ES 3.2 and hardware HEVC decoding on the phone.

WHERE WE STAND

MPEG V3C Immersive Platform

BROUGHT TO STUDY	ON-GOING	OUTCOMES
Support for Haptics over DASH	New Android player	Improved V-PCC rendering in the Unity player
Support for Real-time protocols (RTP, WebRTC) in the player	Support for Meta Quest headsets	Haptic decoder and synthesizer, and V-PCC synthesizer, as source code
Support for an additional synthesizer, such as V-DMC	V3C over DASH streaming to devices	V3C DASH packager and tools to generate V-PCC bitstreams

RELATION WITH SDOS AND INDUSTRY FORA

MPEG (ISO/IEC)

MPEG Contributions m73733 (WG 4) and m73724 (WG 7) in 2025 (MPEG#151) on the release v1.0 of the V3C and haptics immersive platform

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